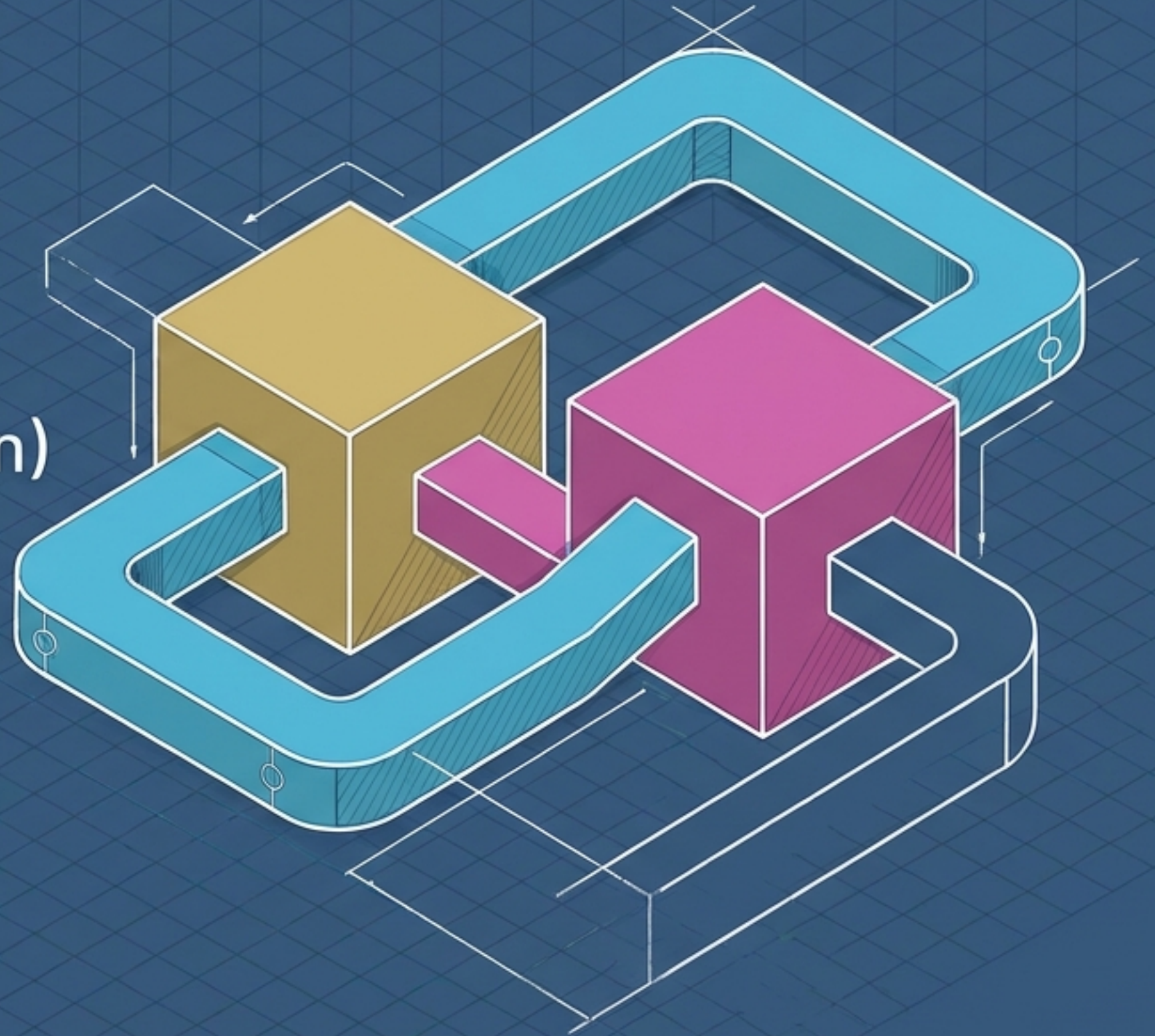


Tying the Data Knot

Predicting Meaningful Connections (V8 Patched Edition)

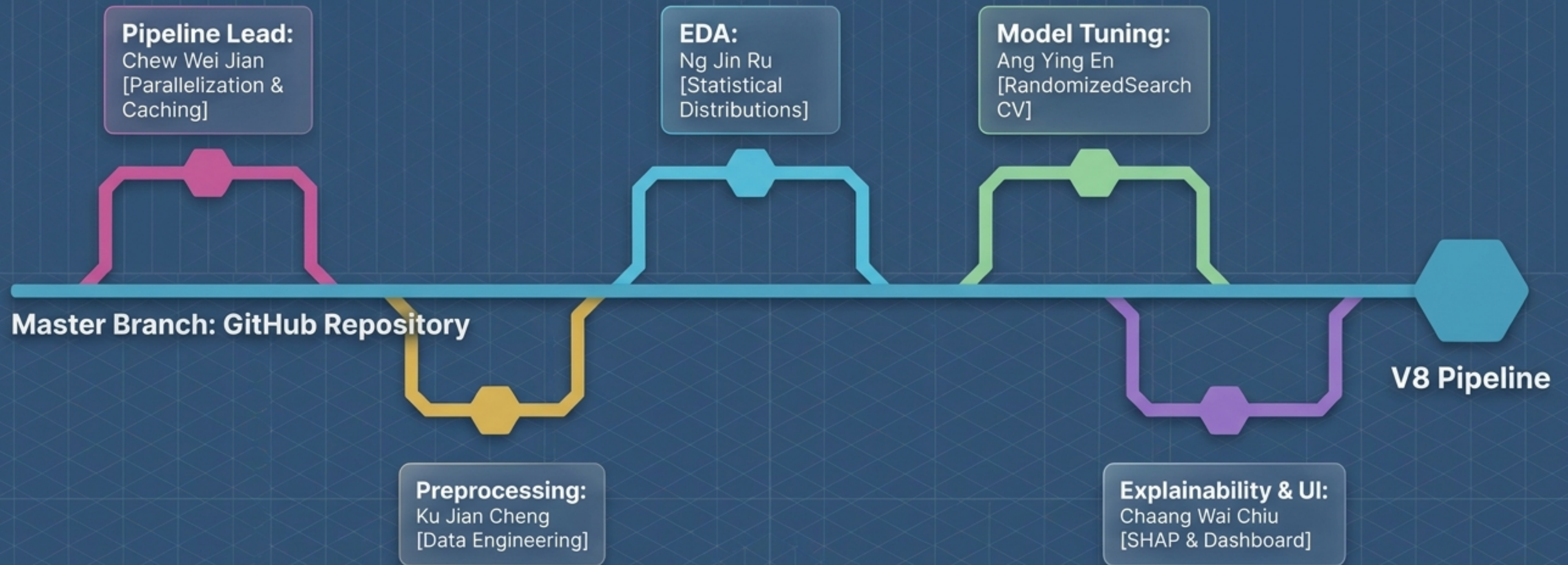
A State-of-the-Art Machine Learning Pipeline for Digital Romance

OCC6 Group 3 | University of Malaya



Parallel Execution and Modular Integration

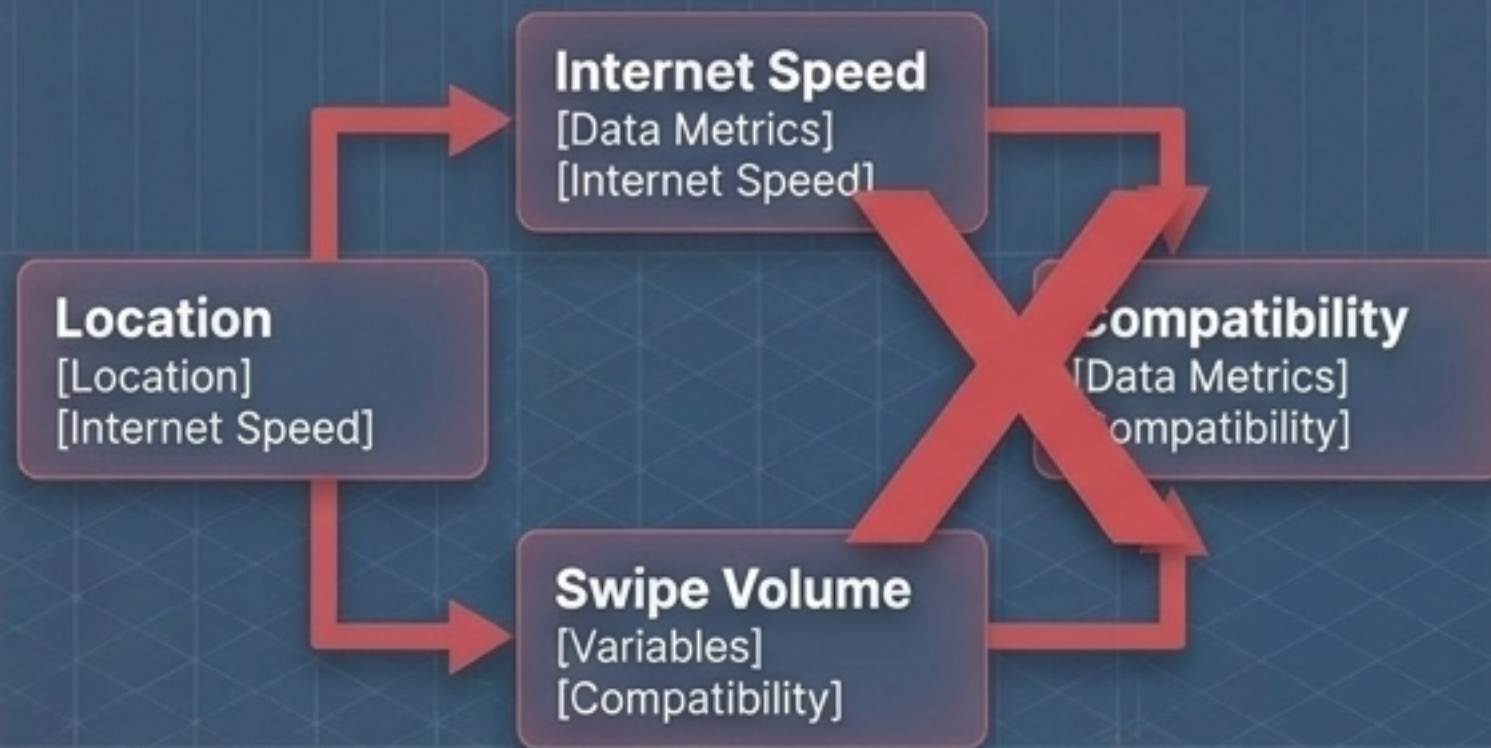
`strict_peer_review_protocol == True`
(No feature selection or baseline merged
to master without validation)



The Illusion of Correlation in Digital Matchmaking

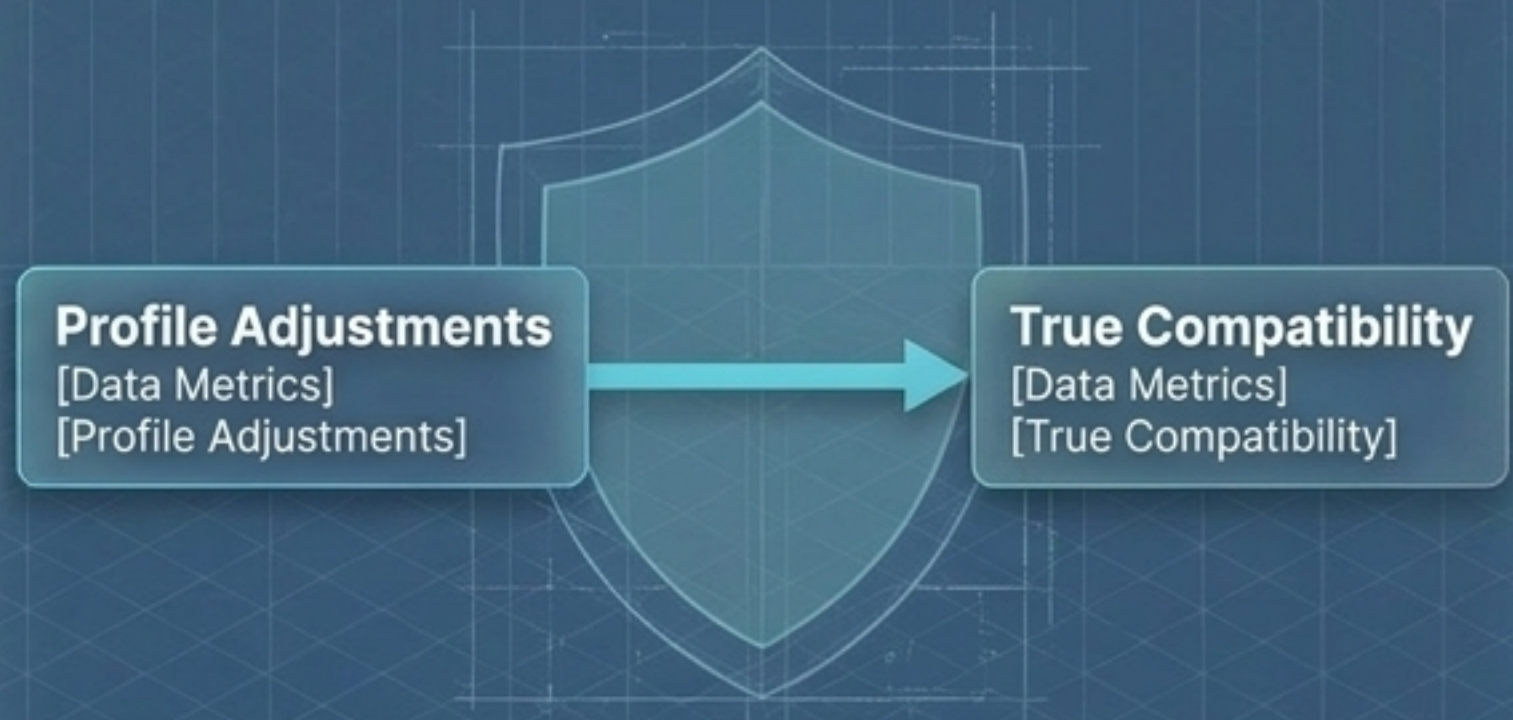
The Problem: High churn and ghosting driven by traditional models conflating spurious correlations (e.g., demographic wealth) with true compatibility.

Predictive ML (The Status Quo)



The Confounding Loop: Static demographic features are heavily mediated by latent confounders.

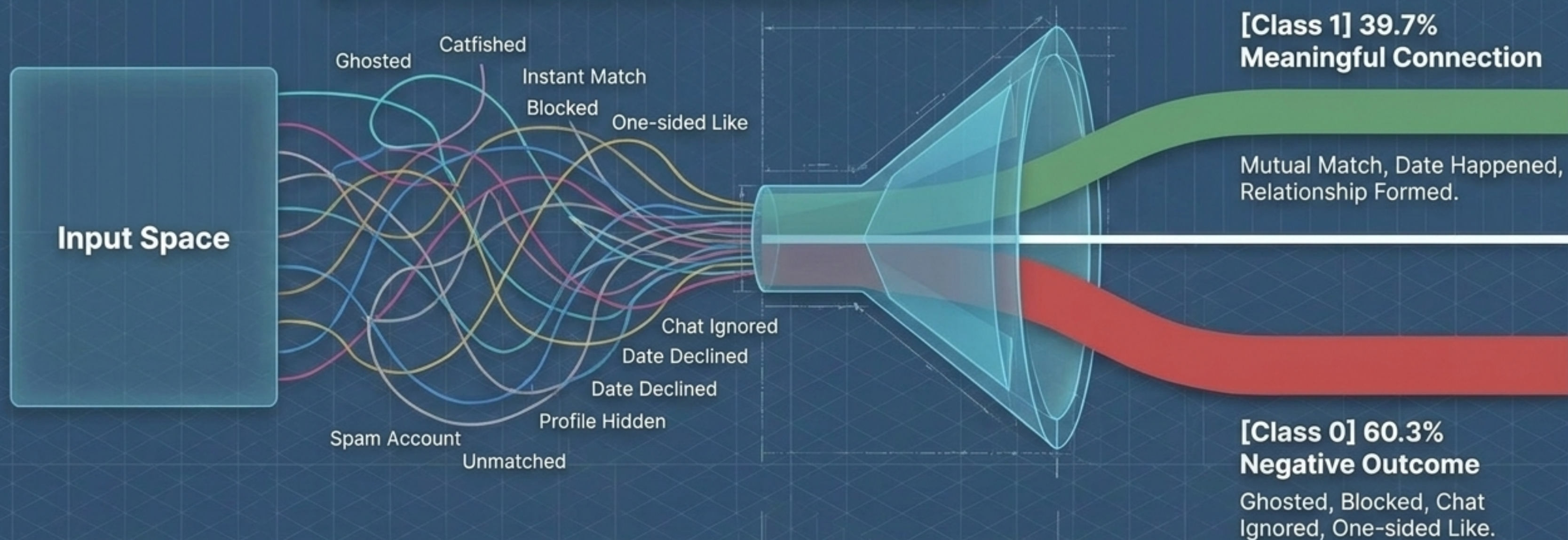
Prescriptive Causal AI (Our Objective)



The Architectural Objective: Transition from prediction to prescription. Build a leakage-free causal audit sandbox to estimate the true interventional probability of a match.

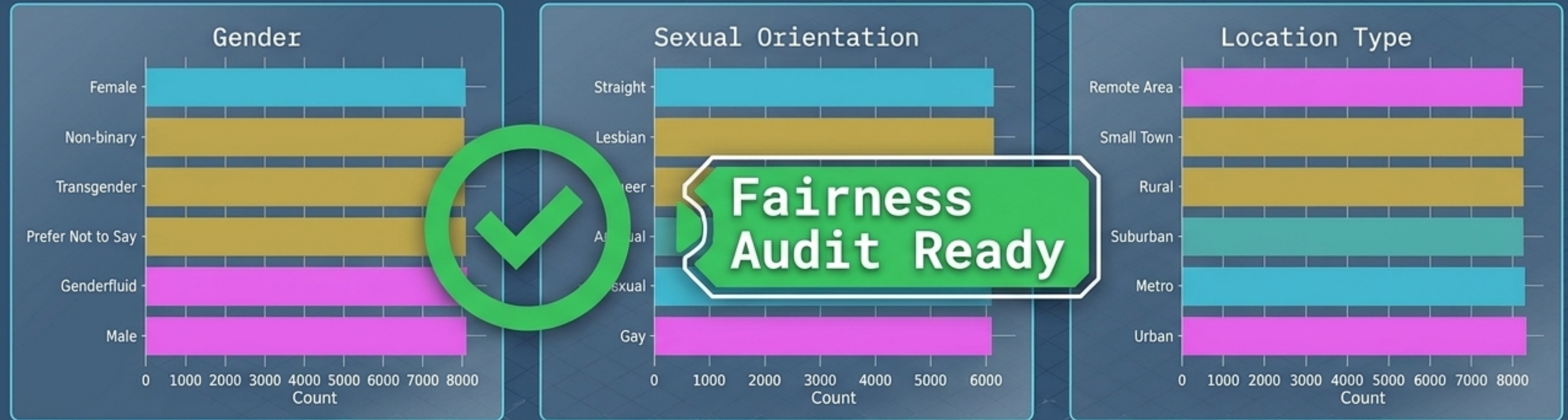
Target Binarization Isolates Meaningful Connections

The Multi-class Problem: 10 raw categorical outcomes.
Warning: Causes extreme label sparsity.



Input Space
50,000 Samples | 25 Behavioral & Demographic Features

Categorical Uniformity Establishes an Ethical Baseline



Observation

Programmatic generation yields perfectly flat, balanced cohorts across Gender, Orientation, and Location.

Algorithmic Advantage

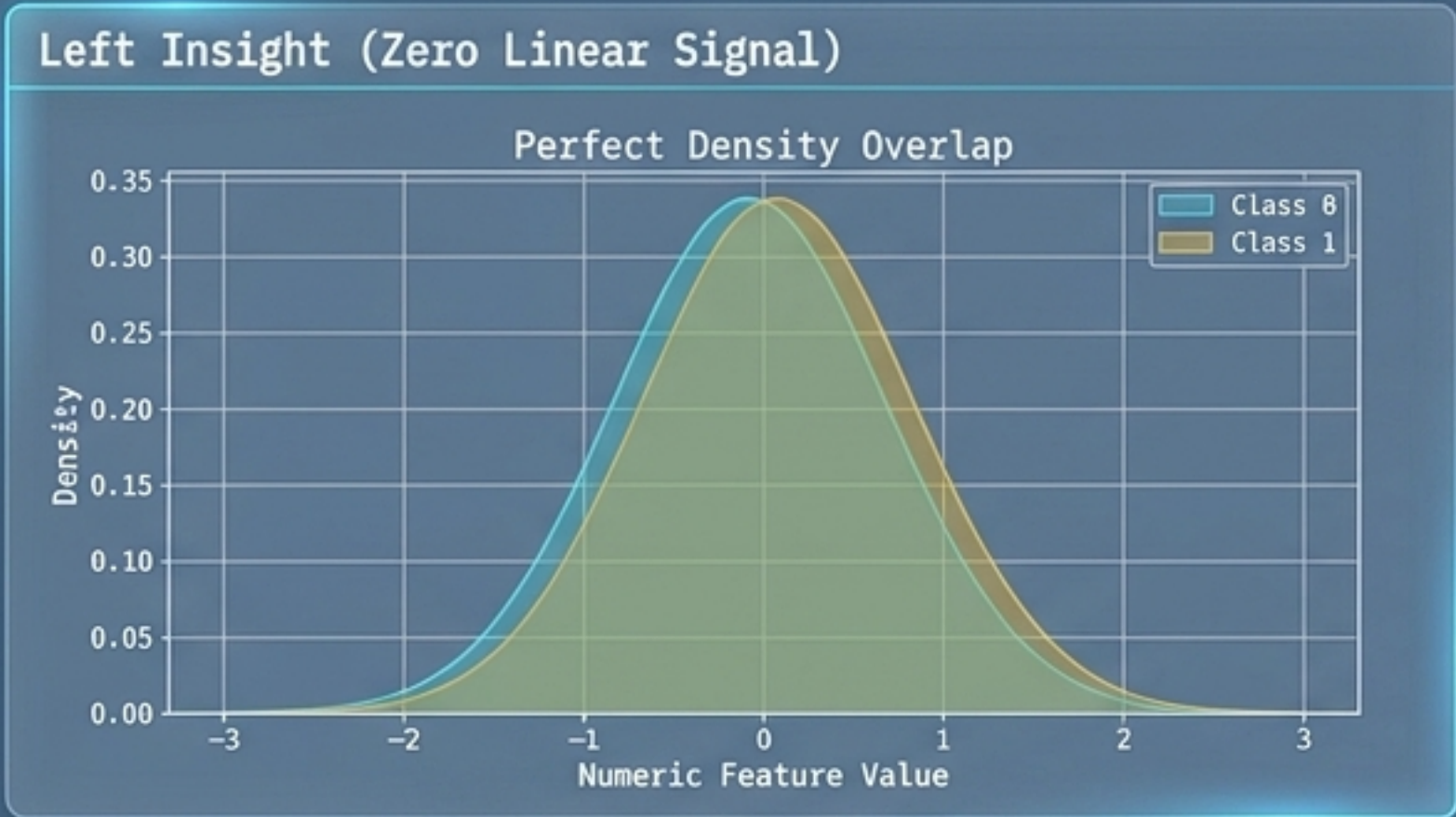
Prevents demographic class dominance in the loss function during gradient descent. No single group can skew the weights.

Ethical Compliance

Establishes a mathematically pristine foundation for downstream Demographic Parity and True Positive Rate (TPR) fairness audits.

Heavy Tails and Zero Linear Signal

Mandate Robust Scaling



Left Insight (Zero Linear Signal): Perfect density overlap between Class 0 and Class 1 metrics. Univariate splits are impossible.



Right Insight: Extreme right-skewed profiles break standard variance margins.

Architectural Mandate:	Action:	Effect:	Constraint:
Rustomkghs	Deploy median-based RobustScaler.	Immunizes distance margins (KNN/SVM) against extreme outliers.	Fitting explicitly deferred post-split to guarantee zero target leakage.